

Sports OS for India (Telangana Focus): Research Findings & Development Specification

TL;DR

- India has no unified "Sports OS": existing apps are fragmented into single-purpose silos — venue booking (Playo, Hudle, KheloMore), cricket-only scoring (CricHeroes, which reports 4 crore+ registered cricketers and scored 4,872,624 matches in 2025 alone), and Western team-admin tools (TeamSnap, which reached 30 million lifetime users in 2025; Spond). None combine clubs/events + multi-sport ball-by-ball scoring + lifelong portable player profiles + government dashboards. **That whitespace is the product thesis.**
- The Telangana timing is ideal: CM Revanth Reddy unveiled the **Telangana Sports Policy 2025** at the state's first Sports Conclave in Hyderabad on **Aug 2, 2025**, backed by a **₹465 crore FY2025-26 sports budget** (₹100 crore more than the prior year), the **Young India Physical Education & Sports University** (targeted to open by end-2028), and a **mandal→district→state** talent pipeline — all of which need exactly the participation data a Sports OS would generate. SATS has even tendered for a web portal + chatbot + WhatsApp bot, signalling appetite for this digital layer.
- Recommended build:** offline-first mobile app (Flutter + local DB + sync engine), per-sport modular scoring engines benchmarked on CricHeroes, a **Glicko-2** player rating with a cross-sport percentile index, a configurable tournament engine (knockout/round-robin/Swiss/groups), and government analytics dashboards. Phase the MVP around clubs+events + cricket + one racket sport, then expand.

Key Findings

The market gap

Existing platforms each solve one slice:

- Venue-booking + social discovery:** Playo, Hudle, KheloMore. Playo monetizes via booking commissions, venue SaaS, premium subscriptions, corporate events, and a "Karma points" loyalty program (earn 3 Karma/booking, 1/game, 50/referral; redeemable up to ~20% of a booking in some cities). Hudle launched **GRIP, "India's first rating system for racket sports"** (1-7 scale from real match results). KheloMore operates in 250+ cities with split-payment and a partner venue-management app. (Dealroom.co)
- Cricket scoring network:** CricHeroes — the dominant grassroots model and the single most important benchmark. It reports **4 crore+ (40M+) registered cricketers** and, per its "Landscape of Grassroots Cricket 2025," **4,872,624 matches scored in 2025 alone** (avg ~13,350/day) across 142 countries, 7,315 cities, with 12,380,543 active players in 2025. Features: free ball-by-ball scoring, wagon wheel/Manhattan/worm graphs, MVP, leaderboards, tournament auto-schedule generator, Smart NRR calculator, live streaming, CricInsights analytics, CricPay for team expenses, and a "world-class cricket profile" with lifetime batting/bowling stats. Freemium + PRO subscription; raised ~\$2.48-2.96M; **FY2023 revenue ₹8.11 crore (₹81,115,000), EBITDA ₹1.62 crore** (Preqin).

- **Team management (international):** TeamSnap — **30 million lifetime users in 2025, 2 million daily active users, 1.6 billion sessions, 19,000+ organisations** (Youth Sports Business Report) — strong scheduling/availability/messaging/payments/registration but a limited free tier and ad-heavy. Spond (fully free incl. events + fundraising; charges only a payment-processing fee), Heja (communication-first), SportsEngine, LeagueApps.

No platform is a true multi-sport "OS" tying together club creation → event/tournament → live multi-sport scoring → lifelong portable profile → talent discovery → government dashboards.

Government alignment (strong tailwind)

National. The Union Cabinet, chaired by PM Narendra Modi, approved the **National Sports Policy 2025 ("Khelo Bharat Niti 2025")** on **July 1, 2025**, superseding the National Sports Policy 2001 and explicitly targeting the **2036 Olympic Games** bid. Vision: "Sports for Nation Building." Five pillars: (1) Excellence on the Global Stage, (2) Sports for Economic Development, (3) Sports for Social Development, (4) Sports as a People's Movement, (5) Integration with NEP 2020. It aims to strengthen sporting culture "from the block level to the national level," serve as a model for state policies, and integrate sport into school/college curricula. Complementary programs: **Khelo India** (grassroots talent ID, competitions, infrastructure across five age groups), **KIRTI** (Khelo India Rising Talent Identification — ages 9–18, AI-based analytics to predict sporting acumen, registration via the **MyBharat** portal, integration with the **Fit India Mobile App**), **Fit India**, **TOPS**. India targets top-10 sporting nation by 2036 and top-5 by 2047.

Telangana. CM A. Revanth Reddy unveiled the **Telangana Sports Policy 2025** at the state's first Sports Conclave (Aug 2, 2025, per All India Radio). Key elements: **₹465 crore FY2025-26 sports budget** (₹100 crore more than the previous year); **Young India Physical Education & Sports University (YIPESU)** on a ~200-acre Hakimpet/Gachibowli campus, PPP model, targeted to open by **end-2028**, with an independent board including Abhinav Bindra, P.T. Usha, Kapil Dev, and Kavya Maran; **14 priority sports** (cricket, hockey, football, basketball, swimming, tennis, badminton, shooting, boxing, wrestling, table tennis, athletics, gymnastics, aquatics); a **Telangana Sports Development Fund (TSDF)** overseen by a **14-member Board of Governors** (co-chaired by Sanjiv Goenka and Upasana Kamineni); and an explicit goal to "bridge the urban-rural divide" and extend training "to every district." **SATS/SATG (Sports Authority of Telangana State)** runs district sports authorities, summer coaching camps (Shaping Young India-2025), CM Cup 2024 (incl. Para Games), and conducts selection trials in **three phases: mandal → district → state**. Hyderabad hosts the Khelo India Games in Nov 2025. SATS has floated a tender for a **web portal + chatbot + WhatsApp bot** — direct evidence of demand for this digital layer. (The Hans India + 4)

The mandal→district→state trial structure and the "every district/village" goal map directly onto a participation-tracking dashboard. This is the government's strategic wedge and the platform's most defensible B2G moat.

Player rating systems

- **ELO** (the user's "EOL"): logistic expected-score model,
$$E = 1 / (1 + 10^{((R_b - R_a) / 400)})$$
; update
$$R' = R + K(S - E)$$
. Simple, but tracks no uncertainty and treats 3 games like

3,000.

- **Glicko / Glicko-2** (Mark Glickman, public domain): adds **Rating Deviation (RD)**, a confidence/standard-deviation measure; a rating of 1500 with RD 50 implies true strength $\sim 1402\text{--}1598$ at 95%) and, in Glicko-2, a **volatility σ** (0.06 default). Rating change is smaller when your RD is low or your opponent's RD is high; RD grows during inactivity. Used by **chess.com** and **lichess**. Reaches accurate estimates faster than Elo with fewer games and reports confidence bounds — ideal for grassroots where players have few recorded games. (arxiv)
- **TrueSkill** (Microsoft): Bayesian Gaussian (μ, σ), starts $N(25, (25/3)^2)$, handles multi-player/free-for-all and teams — relevant for team sports and multi-competitor events. (arxiv)
- Grassroots analogues in-market: CricHeroes leaderboards/MVP, Playo Karma/skill level, Hudle GRIP (1-7).

Recommendation: Glicko-2 per sport as the spine (uncertainty-aware, few-games-friendly, public domain), a TrueSkill-style approach for inferring individual contributions in team sports, and a normalized cross-sport "**Sports OS Index**" (percentile within sport $\times$ recency weighting $\times$ RD confidence) so profiles are comparable across disciplines. Feed sport-specific performance inputs (not just win/loss) into the rating for fairness.

Tournament formats

- **Single elimination / knockout**: N teams, $N-1$ matches; fastest; byes to top seeds.
- **Round robin**: everyone plays everyone; $N(N-1)/2$ matches (10 teams = 45; 16 = 120); use **Berger tables** for optimal scheduling; impractical beyond ~ 12 teams for a single day.
- **Double elimination**: parallel winners + losers brackets; eliminated only after two losses.
- **Swiss**: fixed rounds ($\sim \log_2 N$; 32 players $\rightarrow$ 5 rounds/80 matches vs 496 for round-robin), pair similar records, no rematches, **Buchholz** tiebreaker (sum of opponents' wins); best for large fields with limited time.
- **Group stage + knockout (hybrid)**: pools $\rightarrow$ bracket; the dominant format for large events.
- Points tables, seeding, and NRR-style tiebreakers all required as configurable modules.

Offline-first is non-negotiable for rural India

Grounds in rural Telangana have poor connectivity. Best practice: treat the **local device DB** as **the source of truth**; the UI reads/writes locally first; a **sync engine** queues unsynced records (flagged `isSynced`) and pushes/pulls deltas when online; use client-generated IDs, timestamps, and explicit conflict resolution (last-write-wins baseline, escalate true conflicts to the user). Flutter options: **Isar/Drift/Hive** + **Riverpod** reactive streams; React Native: SQLite + redux-persist + background-fetch.

Payments

Razorpay is the India standard: **UPI accounted for $\sim 85\%$ of India's digital payment volume in H1 2025** (RBI; up from 34% in 2019), so deep UPI support matters more than cards. Use **Razorpay Route** for marketplace split payments (auto-split registration fees to clubs/venues, retain a platform fee), **Subscriptions/UPI AutoPay** for PRO tiers, and **Smart Collect** virtual

accounts; pricing ~2% + 18% GST (~2.36% effective). This directly enables paid event registration with automatic club payouts.

Details

PART 1 — Clubs, Communities & Events

Clubs/Entities. Any individual, community, school, college, or village can create a Club (entity-type flag). Roles: Owner/Admin, Sub-admin, Member, Guest. Invite via phone number, shareable link, or QR. Membership requests/approvals. Sub-groups (age group, team, gender). Multi-language onboarding (Telugu, Hindi, English).

Internal events (free). Club admins create events that push notifications to all members for one-tap registration. Fields: title, sport, venue (map + text), date/time, description, format, capacity/registration limit, waitlist, RSVP (going/maybe/not), skill filter, cost (free by default).

Inter-club challenges. A club/school/college can challenge another; challenge → accept/decline/reschedule → fixture created. Home/away/neutral venue. School-vs-school and college-vs-college leagues.

Event/tournament types. Knockout, round robin (single/double), league, double elimination, Swiss, group stage + knockout. Registration limits, seeding, entry fee (optional).

Paid registration. Optional entry fees via Razorpay UPI; **Razorpay Route** auto-splits to the organizer club minus platform fee; refunds, split payments among teammates (KheloMore/Playo pattern), GST invoices.

Discovery & notifications. Nearby events/tournaments by geolocation, sport, date, skill; club feed; push + **WhatsApp/SMS fallback** (critical where app engagement is low). Benchmark: CricHeroes "Looking" (find teams/players/umpires/grounds), Playo activity hosting, Spond RSVP.

Best practices adopted: Spond's free-everything + payment-processing-only fee (trust builder), TeamSnap availability tracking, Playo reputation badges/peer rank, CricHeroes community roles (scorer/umpire/commentator registration).

PART 2 — Match-Day Tools & Per-Sport Scoring (the heart)

Team formation. Manual drag-and-drop shuffle; captain-pick mode; **AI-balanced teams** using player Glicko-2 ratings + sport-specific stats to minimize the predicted rating differential between sides (bin players by skill, snake-draft, then optimize). Handle late joins/dropouts.

Officiating & live. Assign Scorer, Umpire/Referee, Commentator roles; live score sharing via link (spectators follow ball-by-ball without the app); optional live streaming integration; MVP/Best Player selection; scorer quality score (CricHeroes SQS analogue).

Added (user-missed) match-day modules: umpire/scorer assignment & verification; live score broadcast link; streaming hooks; MVP voting; injury/substitution log; venue/ground management (booking, floodlights, pitch/court status); equipment checklist; weather flag; dispute resolution (both captains confirm result; admin arbitration; scorecard lock after mutual confirmation).

Per-sport scoring specifications

Cricket (benchmark: CricHeroes). *Per-ball model:* innings, over #, ball # within over (6 legal balls/over; wides & no-balls re-bowled and don't count as one of the 6), striker, non-striker, bowler; runs off bat (0-7+ incl. overthrows); **extras** — *wide* (+1, charged to bowler), *no-ball* (+1, free hit in limited-overs, byes/leg-byes recorded under it), *bye* (team total, charged to keeper not bowler), *leg-bye* (team total, not batsman/bowler); **wicket types** — bowled, caught (+catcher), LBW, run-out (+fielder/end, bowler NOT credited), stumped (+keeper), hit-wicket, retired; fall-of-wicket (score+over); **wagon-wheel coordinates** (store as angle+distance or normalized x/y from striker origin); Manhattan/worm/run-rate graphs; UNDO for corrections. *Batting stats:* runs, balls, **SR** = $\text{runs/balls} \times 100$, **avg** = runs/dismissals , 4s/6s, 50s/100s, not-outs. *Bowling:* overs, balls, maidens, runs conceded (bat+wides+no-balls; NOT byes/leg-byes), wickets (excl. run-outs), **economy** = runs/overs , avg, strike rate. *Fielding:* catches, stumpings, run-outs. **NRR** = $(\text{runs scored/overs faced}) - (\text{runs conceded/overs bowled})$, aggregated across the whole tournament, to 3 decimals; overs decimalized by balls/6 (47.2 overs = 47.333, NOT 47.4); if a team is bowled out, use its **full allotted overs**. Formats: T10/T20/ODI/custom overs, Test/multi-day. Officials: 2 on-field umpires, scorer, optional third umpire. (Sports Rules + 3)

Football (soccer). Score = goals. Events: goal (+scorer, +assist, minute), own goal, penalty (scored/missed), yellow/red card, substitution, foul, corner, offside, shot (on/off target), save. Player stats: goals, assists, minutes, cards, saves (GK), clean sheets, shots, pass completion (advanced). Formats: full 90' (2×45), small-sided 5-a-side/7-a-side with configurable halves; league points 3/1/0; knockout with extra time + penalty shootout. Officials: referee, assistants.

Badminton. Rally-point. Current widespread standard: **21-point, best of 3, win by 2, cap 30** (29-29 → 30 wins). **Note: the BWF approved a 3×15 system for elite play (best of 3 to 15, cap 21, change ends/interval at 8) reported for 2026** — make target score and best-of **configurable per event**. Serve court determined by even/odd server score. Events: point (+winner, +rally, serve side), fault type, interval (11 or 8), change of ends. Stats: points won on serve/receive, games won, longest rally. Singles & doubles. Officials: umpire, service judge, line judges.

Volleyball. Rally-point; best of 5 sets; sets to 25 (win by 2, no cap); deciding 5th set to 15 (win by 2); switch every rally. Events: point (+type: attack/block/ace/opponent error), rotation, substitution, timeout, faults (net touch/double-hit/foot-fault/four-hits). Stats: points, aces, blocks, kills, digs, assists. Officials: 1st/2nd referee, scorer, line judges.

Kabaddi. Two 20-min halves (international 2×20; 5-min break). 7 on court + 5 subs. Raider scores **touch points** (1 per defender tagged) + **bonus point** (touching bonus line when 6+ defenders, in addition to touch); defenders score **tackle point** (1) for stopping the raider; **super tackle** (2 pts when ≤3 defenders on court); **super raid** (3+ points in one raid); **all-out** = +2 bonus and revives whole team; **do-or-die raid** after 2 empty raids by a team. 30-second raid clock. Events: raid (raider, defenders out, type, bonus), tackle (defenders involved, super), empty/do-or-die, all-out, revival, technical points. Stats: raid points, tackle points, super raids/tackles, "Super 10" (10 raid pts), "High 5" (5 tackle pts). Officials: referee, 2 umpires, scorer, TV umpire.

Kho-Kho (modern / Ultimate Kho Kho). 2 innings; each innings = attack + defense turn of **7 minutes each** (traditional international was 9); defenders enter in **batches of 3**. Scoring (UKK Season-2 simplified — treat as **configurable**): regular tag/touch **2 pts**, **Pole Dive 2 pts**, **Sky Dive 2 pts**, **Dream Run** +1 after 3 min survival then +1 per further 30s; all-out ("Lona") bonus. UKK innovations: **Wazir** (multi-directional runner, like an impact player) and powerplay. Events: kho given, tag out (+skill type: regular/pole-dive/sky-dive/self-out), batch entry/exit, dream-run timers, fouls, cards, powerplay, all-out, reviews. Stats: touch/sky-dive/pole-dive/dream-run points, khos, time survived. Officials (UKK, ~10): chief/TV referee, on-field referee, 2 umpires, 2 pole umpires, scorer + assistant, 2 timekeepers. **Flag: point values changed between seasons — make configurable per league/season.** (Ultimatekhokho + 4)

Basketball. Points 1 (FT)/2/3. Events: made/missed shot (+type, +coordinates for shot chart), rebound (off/def), assist, steal, block, turnover, foul (personal/technical), substitution, timeout, quarter. Stats: points, rebounds, assists, steals, blocks, turnovers, FG%/3P%/FT%, +/–, PER (advanced). Formats: 4 quarters (configurable minutes); 3×3 half-court to 21 or 10 min. Officials: 2–3 referees, scorer, timekeeper (shot clock).

Tennis. Points **15/30/40**, deuce → advantage (Ad-In/Ad-Out) → game (win by 2); or **no-ad** (next point wins). Set = first to **6 games, win by 2**; at 6–6 a **tiebreak** to 7 (win by 2; first server serves 1 point, then alternate serve every 2 points; change ends every 6 points). Match: **best of 3** (win 2) standard; **best of 5** for men's Grand Slams; a 10-point **match tiebreak** replaces the deciding set in many formats — make best-of **configurable**. Change ends after odd games. Events: point (+server, +serve outcome), ace, double fault, break point (faced/saved/converted), winner, unforced error. Stats: aces, DFs, 1st-serve %, 1st/2nd-serve points won %, break points, winners, UEs, games held/broken. Officials: chair umpire, line judges. (Wilson Sporting Goods)

Table Tennis. Game to **11, win by 2** (no cap; 10–10 continues to 12–10, 13–11...); serve alternates **every 2 points** (every 1 point at 10–10 deuce); **best of 5** (to 3) club / **best of 7** (to 4) ITTF/Olympic — configurable. Change ends each game; in the deciding game switch ends at 5. Serve: ≥16 cm near-vertical toss, from behind the end line, ball visible; net-clip serve = **let** (unlimited). Doubles ABCD rotation. Events: point (+server, +winner, +serve #), let, net/edge, expedite trigger. Stats: service/receive points won %, games won, deuce record, longest rally. Officials: umpire, assistant umpire. (PongFit)

Hockey (field). Score = goals (field goal / penalty corner / penalty stroke). Events: goal (+type, +scorer, +assist), penalty corner, penalty stroke, green/yellow/red card, substitution, quarter. Stats: goals, assists, cards, PC conversion, saves (GK). Formats: 4×15-min quarters. Officials: 2 umpires, scorer, timekeeper.

Chess. Result 1/0.5/0 per game; formats: classical/rapid/blitz/bullet (configurable time control); Swiss or round-robin tournaments; Buchholz/Sonneborn-Berger tiebreaks. Events: moves (optional PGN capture), result, time control. Rating: **Glicko-2 per time control** (as chess.com/lichess). Officials: arbiter.

Athletics / track & field. Event-based, not head-to-head: record times (track, to 0.01s or photo-finish), distances/heights (field). Events: heats → finals, lanes, false starts, wind reading,

attempts (field). Stats: personal bests, season bests, rankings by event/age group. Officials: starter, timekeepers/photo-finish, field judges.

Carrom. Points per pocketed coin + queen (with cover); board = 29 points; best-of series.

Events: coin pocketed, queen, foul, board win. Stats: points, boards won, queens. Configurable to state association rules.

PART 3 — Post-Match, Memories & Lifelong Profiles

Post-match: declare winner, awards/MVP (voted or stat-computed), share scorecard link, upload photos/videos → auto-compiled "**Memories**" per match/tournament (timeline + highlights). Result confirmation by both captains locks the scorecard (dispute resolution).

Lifelong portable profiles. One identity (phone number + optional Aadhaar/One-ID linkage) that persists across clubs, cities, and decades. A player who played for a Hyderabad community club, then Delhi University, then a corporate team, has a single continuous career: all matches, per-sport stats, ratings over time, awards, teams, and memories. **Verification tiers:** self-reported → scorer-verified → tournament/association-verified (like CricHeroes association integration, chess.com continuous history, Strava lifetime activity log). A career-aggregation service rolls up ball/point-level events into lifetime stats. **Profile portability across entities is the core moat** vs. single-club apps.

Talent discovery. Scout/selector/academy role with search & filters (sport, age group, district/mandal, rating percentile, verified stats, recent form). Shortlists, trial invitations, watchlists. Feeds directly into KIRTI-style AI talent prediction and SAI/SATS mandal→district→state trials. Consent + minor-safeguarding controls (guardian approval for under-18 visibility).

PART 4 — Government / Telangana Dashboards

Administrator dashboards (state / district / mandal / village) tracking: participation counts by sport, area, age group, gender, disability status; number of clubs, events, tournaments, matches; infrastructure/venue utilization; talent pipeline (identified → trialed → selected); women's & para participation (policy KPIs). Filters map to Khelo India age groups and Telangana's 14 priority sports. Export/API for SATS, School Games Federation of India, university sports, Khelo India/MyBharat, and Fit India. Public transparency views. This turns grassroots activity into the exact evidence base the Sports Policy 2025 and TSDF need — a natural PPP/CSR onboarding pitch aligned with SATS's own portal/chatbot tender.

Recommended Technical Architecture

- **Client: Flutter** (single codebase, strong offline-first story with Isar/Drift + Riverpod) — recommended over React Native for scoring-engine performance and offline maturity; either is viable. Mobile-first, low-end-device optimized, Telugu/Hindi/English i18n from day one.
- **Offline-first scoring:** local DB as source of truth; **append-only event log per match** (ball/point/raid events) with client-generated UUIDs + timestamps; background sync engine with delta push/pull; conflict resolution last-write-wins with a per-match single-

scorer lock to avoid conflicts; scorecard fully reconstructable from the event log (**event sourcing**).

- **Backend:** API (Node/NestJS or Go) + **PostgreSQL** (relational core) + Redis; **WebSockets** for live score broadcast; object storage (S3-compatible) for media/memories; push (FCM) + WhatsApp/SMS gateway; CDN.
- **Real-time:** publish each synced event to spectators via WebSocket channels per match; lightweight public web viewer (no install).
- **Payments:** Razorpay (UPI, Route splits, Subscriptions).
- **Ratings service:** Glicko-2 recompute per sport per rating period; cross-sport index as a materialized view.
- **Scalability:** stateless API horizontal scaling; read replicas; partition event tables by match; queues for sync + notifications.

Core data model (entities & relationships)

- **User** (id, phone, name, DOB, gender, district/mandal, languages, verification tier, One-ID link)
- **Entity/Club** (id, type: individual/community/school/college/village, name, location, admins)
- **Membership** (user_id, club_id, role, status, subgroup)
- **Event** (id, club_id, sport, type, venue, datetime, capacity, fee, format)
- **Tournament** (id, event_id, format, seeding, points-config, tiebreak-config)
- **Team** (id, tournament/event_id, name, players[])
- **Match** (id, event/tournament_id, sport, teams/players, venue, scorer_id, status, result, MVP)
- **MatchEvent** (id, match_id, seq, type, payload JSON — ball/point/raid/goal, player refs, coordinates, timestamp, isSynced) — polymorphic per sport
- **Innings/Set/Game/Period** (match_id, structure per sport)
- **PlayerMatchStats** (match_id, user_id, sport-specific stat blob) → aggregated to **PlayerCareerStats** (user_id, sport, lifetime aggregates)
- **Rating** (user_id, sport, glicko_rating, RD, volatility, updated_at) + **CrossSportIndex**
- **Payment** (event_id, user_id, amount, split, status)
- **Media/Memory** (match/tournament_id, url, type, uploader)
- **Dashboard aggregates** (area, sport, age_group, period → counts), materialized for government views

Recommendations

Stage 1 — MVP (0-4 months): Clubs + Events + first scoring engines. Build club creation/membership/invites, internal + inter-club events, notifications, and free registration. Ship **cricket** (highest demand, clearest benchmark) + **one racket sport (badminton)** scoring, offline-first, with live share links and basic profiles. Launch with 2-3 Hyderabad

communities/colleges as design partners. Benchmark to beat: CricHeroes for cricket UX.

Threshold to proceed: $\geq 1,000$ matches scored and $\geq 60\%$ scorer retention across 3 clubs.

Stage 2 — Scale sports + tournaments + payments (4-8 months). Add football, volleyball, kabaddi, basketball, table tennis, tennis, kho-kho, chess. Full tournament engine (all formats + auto-scheduler + points tables/NRR). Razorpay paid registration + Route splits. Glicko-2 ratings + AI team balancing. **Threshold:** first 20 paid tournaments settled cleanly; rating system stable (RD converging).

Stage 3 — Lifelong profiles + talent discovery + memories (8-12 months). Career aggregation, verification tiers, portability, scout/academy role, cross-sport index, memories/media. Pilot talent-discovery with one academy/SATS district. **Threshold:** verified profiles portable across ≥ 2 clubs for real users.

Stage 4 — Government dashboards + integrations (12-18 months). District/mandal/state dashboards; MoUs with SATS, SGFI, Khelo India/MyBharat; PPP/CSR funding via TSDF. Position as the participation-data backbone for Sports Policy 2025. **Threshold:** ≥ 1 district signs on; dashboard reflects real mandal-level participation.

Monetization: keep scoring/clubs **free forever** (the CricHeroes/Spond trust model). Revenue from: (1) tournament-organizer PRO tier + payment take-rate via Route; (2) venue-booking commissions + venue SaaS (Playo model); (3) academy/coaching marketplace; (4) **B2G/CSR dashboard licensing** to the state/SATS; (5) sponsorships/branded tournaments; (6) merchandise/streaming add-ons.

Cross-cutting: Telugu-first localization; WhatsApp/SMS as a notification fallback; minor-safeguarding and data-consent (especially for talent visibility of under-18s); DPDP Act compliance for personal/athlete data.

Caveats

- **Kho-kho point values vary by league/season** (UKK Season 2 simplified 2/2/2 vs. other KKFI-aligned docs listing 3 pts for sky/pole dives and a 2 min 30 sec dream-run threshold). Build all point values as **configurable per league/season**, not hard-coded.
- **Badminton is mid-transition:** the BWF approved a 3×15 elite scoring system (reported for 2026); 21-point remains widespread at grassroots. Make target score and best-of configurable per event.
- **CricHeroes user figures are self-reported marketing** (variously 40M+/49M+); financials from third-party aggregators (Preqin/Tracxn/Getlatka) vary and may be dated — treat as indicative, not audited.
- **Telangana Sports Policy 2025** full text and some university details were reported via news/press; specific figures (₹465 crore budget, ~₹1,000 crore intent, 2028 opening) come from statements and budget announcements and may evolve on implementation.
- **Wagon-wheel coordinate schema** is described functionally by CricHeroes, not as a published spec — coordinate storage format is an implementation choice (recommend angle+distance or normalized x/y from the striker origin).

- **Rating-system critique:** Elo/Glicko/TrueSkill essentially track weighted win/loss, not margin of victory or opponent context beyond rating; incorporate sport-specific performance inputs into the rating for grassroots fairness.
 - **"One Nation One Sport ID" / national athlete registry:** KIRTI/MyBharat registration is the closest live mechanism; a single formal national athlete ID was not confirmed in sources — design the profile identity to be linkable if/when such a registry formalizes.
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This report is structured to be converted directly into a `claude.md` specification: Parts 1-4 map to feature modules, the per-sport section provides scoring-engine requirements, and the data model / architecture section provides the build scaffold. All hedged claims that could be pinned to a named source have been; remaining caveats are flagged explicitly above so they can be encoded as configuration rather than assumptions.